Pengfei Ma

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Education

Sichuan University 2023 – 2027
B.Eng. in Computer Science and Technology *Chengdu, China*

GPA 3.67 / 4.0
Average Score 87.8 / 100
Comprehensive Ranking **17 / 319**

Publications & Patents

[1] GhostODE: A Lightweight Network Optimized by Ordinary Differential Equation [Link]
- **Pengfei Ma**, Le Yu, Lei Zhang, Zhang Yi, Tao He
- Accepted, **Neurocomputing 2026**

[2] Boundary-Aware Test-Time Adaptation for Zero-Shot Medical Image Segmentation [Link]
- Chenlin Xu, Lei Zhang, Lituan Wang, Xinyu Pu, **Pengfei Ma**, Guangwu Qian, Zizhou Wang, Yan Wang
- Accepted, **International Conference on Intelligent Computing 2026 (Oral)**

[3] Lightweight Image Segmentation Method and System Based on NeuralODE Neural Network Architecture
- **Pengfei Ma**, Tao He, Jianlong Xiong, Yue Cao
- Granted, China National Invention Patent, No. CN121999231B

[4] A Dynamic Prompt-Constrained and Edge-Anchor-Guided Image Segmentation Method, Apparatus, Storage Medium, and Electronic Device Based on the SAM Model
- Chenlin Xu, Lituan Wang, Lei Zhang, **Pengfei Ma**, Xinyu Pu, Yuxin Xie, Yitao Zhu
- Granted, China National Invention Patent, No. CN120279276B

[5] MDT-Based Diet Recommendation Method, System, Device, and Medium for Chronic Kidney Disease
- Yuheng He, **Pengfei Ma**, Shutong Cheng, Siyang Li, Yifan Li, Wenqiang Lei, Xi Chen, Xiaoyan Yang
- Under Substantive Examination, China National Invention Patent, Publication No. CN122552039A

Research Experience

GhostODE: Lightweight Network via Neural ODE

Advisors: Prof. Tao He *2025 – 2026*

- Identified the significant local discretization error of first-order Euler solvers in lightweight Neural ODEs; proposed a Direct Differentiation Heun (DDH) method, improving integration accuracy to second-order with a marginal $< 7\%$ FLOPs increase.
- Designed a stage-selective Ghost compression strategy, replacing parameter-heavy pointwise convolutions in deep stages (Stages 4-5) with Ghost modules, successfully reducing total parameters by 86.4% vs. MobileNetV1 and 56.3% vs. MobileODE.
- Adopted Decoupled Knowledge Distillation (DKD) as an auxiliary training strategy to compensate for capacity loss, boosting the compact model's performance across classification, object detection, and semantic segmentation tasks.
- Independently executed the full research pipeline from algorithm design to edge-device deployment, resulting in a first-author *Neurocomputing* paper and a granted national invention patent.

SAM-Based Medical Image Segmentation

Advisors: Prof. Lituan Wang *2024 – 2025*

- Proposed an efficient SAM fine-tuning framework leveraging weakly annotated labels (points/scribbles/boxes) while preserving segmentation consistency, addressing the prohibitive cost of pixel-level annotations.
- Designed and implemented the core model architecture and boundary-aware attention module; conducted extensive ablation studies and cross-modality experiments to validate model robustness.
- Resulted in a conference publication and a granted invention patent.

MDT Multi-Agent Dietary System for Kidney Diseases

Advisors: Prof. Wenqiang Lei & Prof. Xi Chen *2025 – 2026*

- Developed a Multi-Disciplinary-Team (MDT) multi-agent architecture combining LLM-based clinical reasoning and Retrieval-Augmented Generation (RAG) over nephrology guidelines for personalized diet planning.
- Engineered the agent orchestration and retrieval-augmented dialogue pipeline, successfully deploying the full system on the Huawei Kunpeng ARM platform.

Awards

National Encouragement scholarship *2026.09*

Comprehensive Second-Class Scholarship, Sichuan University (9/320 that year) *2025.10*

Comprehensive Third-Class Scholarship, Sichuan University *2026.09*

Single-Item Second-Class Scholarship, Sichuan University *2024.11*

School Outstanding Student, Sichuan University *2024.10*

Anheng Spark Encouragement Scholarship *2024.12*

First Prize, Central and Southwest Regional — National University Student IoT Design Competition *2026.08*

Silver Medal, Higher-Education Track — Huawei Kunpeng Innovation Contest 2025, Southwest Regional *2025*

Provincial Third Prize — 9th Huawei ICT Competition (China), Innovation Track, Sichuan Province *2025.04*

Provincial Second Prize — 2024 RAICOM Robot Developer Contest, Sichuan Big-Data Project *2024.11*

Language & Skills

- **Languages:** Mandarin (Native), English.
- **Programming & Tools:** Python, PyTorch, C/C++, Linux, Git, Docker, MindSpore.
- **Hobbies:** Reading, Hiking, Watercolor Painting.

Leadership, Service & Public Engagement

Open-Source Hardware Association, Sichuan University

Head of Project Department | Awarded Top 10 Student Clubs of SCU *2024 – 2025*

- Led the Project Department for one full academic year — organized technical workshops, mentored junior members on embedded systems and rapid-prototyping projects, organized competitions, and promoted knowledge dissemination. The club was selected as one of the **Top 10 Student Clubs of Sichuan University** for the 2024–2025 academic year.

Xinhua News Agency

Interviewee — Telephone interview on the Micro-major Program *2026.08*

- Discussed the curriculum experience, interdisciplinary training, and practical outcomes of the micro-major program from a student perspective.